Table 6 Overview of categorical emotion implementations.

Categorical emotion	**Paper**
Joy/happy	(Gadanho and Hallam, 2001) (Von Haugwitz et al, 2012) (Ficocelli et al, 2015) (Tanaka et al, 2004) (Goerke, 2006) (Yu et al, 2015) (Lee-Johnson et al, 2010) (Williams et al, 2015) (Hasson et al, 2011) (Moussa and Magnenat-Thalmann, 2013) (Salichs and Malfaz, 2012) (Cos et al, 2013) (Moerland et al, 2016) (Jacobs et al, 2014) (Lahnstein, 2005) (Shi et al, 2012) (El-Nasr et al, 2000) (Kubota and Wakisaka, 2010)
Sad/unhappy/distress	(Gadanho and Hallam, 2001) (Von Haugwitz et al, 2012) (Ficocelli et al, 2015) (Tanaka et al, 2004) (Yu et al, 2015) (Lee-Johnson et al, 2010) (Moussa and Magnenat-Thalmann, 2013) (Salichs and Malfaz, 2012) (Moerland et al, 2016) (Jacobs et al, 2014) (Lahnstein, 2005) (El-Nasr et al, 2000) (Kubota and Wakisaka, 2010)
Fear	(Gadanho and Hallam, 2001) (Von Haugwitz et al, 2012) (Tanaka et al, 2004) (Goerke, 2006) (Yu et al, 2015) (Lee-Johnson et al, 2010) (Williams et al, 2015) (Salichs and Malfaz, 2012) (Moerland et al, 2016) (Jacobs et al, 2014) (Matsuda et al, 2011) (Shi et al, 2012) (El-Nasr et al, 2000) (Kubota and Wakisaka, 2010)
Anger	(Gadanho and Hallam, 2001) (Von Haugwitz et al, 2012) (Ficocelli et al, 2015) (Tanaka et al, 2004) (Goerke, 2006) (Yu et al, 2015) (Hasson et al, 2011) (Moussa and Magnenat-Thalmann, 2013) (Shi et al, 2012) (El-Nasr et al, 2000) (Kubota and Wakisaka, 2010)
Surprise	(Von Haugwitz et al, 2012) (Tanaka et al, 2004) (Lee-Johnson et al, 2010)
Hope	(Moerland et al, 2016) (Jacobs et al, 2014) (Lahnstein, 2005) (El-Nasr et al, 2000)
Frustration	(Hasson et al, 2011) (Huang et al, 2012) (Tsankova, 2002)

plemented dimension is valence. Not surprisingly, valence has mostly been derived from reward-based elicitation methods (Broekens et al, 2007a) (Ahn and Picard, 2006) (Zhang and Liu, 2009) (Obayashi et al, 2012) (Hogewoning et al, 2007). It is also connected to a few extrinsic homeostasis papers (Coutinho et al, 2005) (Gadanho, 2003), but then it is referred to as 'well-being'. Although this is not completely the same concept, we group these together here for clarity.

Following the dimensional emotion models of Russell and Barrett (1999) introduced in Section 2.1, the second most implemented dimension is arousal. Arousal has been connected to extrinsic homeostatic dimensions (e.g. pain and overall well-being (Coutinho et al, 2005)), appraisal-like dimensions (e.g. continuation of incoming stimulus (Kuremoto et al, 2013)), and a few hard-wired implementations (Ayesh, 2004) (Guojiang et al, 2010). Note that some do not use the term arousal but refer to similar concepts, e.g. relaxation (Coutinho et al, 2005) and restlessness (Ayesh, 2004). The only paper to extend the valence-arousal space is by Hoey et al (2013), who also include control.

In general, the dimensional emotion models seem somewhat under-represented compared to the categorical emotion implementations. Although the implementation for valence shows some consistency among papers, there is more difficulty to specify arousal or different emotion dimensions. Nevertheless, the continuous nature of dimensional emotion models remains appealing from an engineering perspective. A possible benefit is the identification of a desirable target area in affective space, towards which the agent aims to progress (Guojiang et al, 2010).

Table 7 Overview of four categorical emotion (columns) elicitations for different papers (rows). The text in each cell specifies the elicitation condition. We observe different categories of emotion elicitation, i.e. homeostatic (blue, Section 4.1), appraisal (green, 4.2) and value-based (red, 4.3). We see how single emotions are connected to different elicitation methods (multiple colours in single column) and how single papers use different elicitation methods (multiple colours in single row).

	Happy/Joy	**Sad/Distress**	**Fear**	**Anger**
(Gadanho and Hallam, 1998)	High energy	Low energy	Pain	High restlessness (low progress)
(Goerke, 2006)	All drives low	-	Homesick & low energy	Hunger & homesick & high energy
(Kim and Kwon, 2010)	Goal achievement	No goal achievement	Pain	No progress
(Williams et al, 2015)	Progress & control & low pain	-	Pain & novelty	-
(Salichs and Malfaz, 2012)	Positive delta well-being	Negative delta well-being	Worst historical Q(s,a)	-
(Moerland et al, 2016)	Positive TD	Negative TD	Anticipated negative TD	-
(Shi et al, 2012)	Increasing positive reward	-	Increasing negative reward	Decreasing positive reward
(Yu et al, 2015)	High well-being	Egoistic agent & low well-being	Agent defects & others co-orperate	Agent co-orperates & others defect

Table 8 Overview of dimensional emotion implementations.

Dimensional emotion	**Paper**
Valence	(Kuremoto et al, 2013) (Ahn and Picard, 2006) (Zhang and Liu, 2009) (Broekens et al, 2007a) (Broekens, 2007) (Obayashi et al, 2012) (Hogewoning et al, 2007) (Hoey et al, 2013) (Guojiang et al, 2010) (Coutinho et al, 2005)
Arousal	(Kuremoto et al, 2013) (Obayashi et al, 2012) (Ayesh, 2004) (Hoey et al, 2013) (Guojiang et al, 2010) (Coutinho et al, 2005)
Control	(Hoey et al, 2013)

6 Emotion function

We now discuss the ways in which emotions may influence the learning loop. It turns out emotions have been implicated with all main aspects of this loop: Reward (6.1), State (6.2), Adaptive Critic (6.3) and Actor (6.4). Finally, emotion has also been studied as an epiphenomenon, i.e. without any effect on the learning loop, but for example to communicate the learning/behavioural process to other social companions (6.5). These categories are visualized in Figure 1 (labels I-V). Note that this Section introduces the ways in which emotion may influence the RL loop

on a conceptual level. We summarize the resulting effect, for example on learning efficiency, in Section 7.

6.1 Reward modification

A large group of emotional RL implementations use emotions to modify the reward function. These approaches add an additive term to the reward function that relies on emotions (we have only encountered additive specifications). The reward function is given by

$$r_t = \tilde{r}_t + r_t^{\triangle} \tag{15}$$

where $\tilde{r}(t)$ denotes the external reward function and $r^{\triangle}(t)$ an internal reward based on emotional mechanisms. In the RL community, Eq. 15 is known as *reward shaping* (Ng et al, 1999). The internal reward can be targeted at maximizing positive emotions, but is also frequently associated to homeostatic variables or appraisal dimensions (see Sections 4.1 and 4.2 for elicitation). However, the general underlying principle usually remains that agents seek to maximize positive emotions and minimize negative emotions.

Homeostasis. For homeostatic systems the reward becomes dependent on the current state of the internal homeostatic variables. Some implementations use the difference in overall well-being,

$$r_t^{\triangle} = W_t - W_{t-1} = D_{t-1} - D_t \tag{16}$$

where the step from well-being W to overall drive D naturally follows from Equation (7). In this specification, the acquisition of food does not provide any reward if the associated homeostatic variable (e.g. energy/sugar level) is already satiated. Implementations of the above idea can be found in (Gadanho and Hallam, 2001), (Salichs and Malfaz, 2012) and (Cos et al, 2013). Variants of this have focussed on using positive emotions (instead of well-being) as the reinforcement learning signal, e.g. in (Gadanho and Hallam, 1998) and (Goerke, 2006).

Appraisal-based. Similar ideas are used for appraisal-based reward modifications. Some examples of appraisal dimension specifications were discussed in Section 4.2, with some formal examples in Equations 9-13. Appraisal dimensions are related to generic concepts of the agent history (novelty, recency, consistency of observations with world model) and expectations with respect to the goal (motivational relevance, intrinsic pleasantness). Several studies in the intrinsically motivated reinforcement learning literature have identified the learning and survival benefit of these dimensions (Oudeyer and Kaplan, 2007; Oudeyer et al, 2007). Some authors therefore took appraisal theory as an inspiration to develop intrinsic motivation features.

Specifications in this direction therefore usually take the following form:

$$r_t^{\triangle} = \sum_{j=1}^{J} \phi_j \zeta_j(g_t) \tag{17}$$